

Q1. For the system

$$G(s) = \frac{1}{(s+2)(s+4)}.$$

- (a) Find analytical expressions for the magnitude and phase responses.
- (b) Also make plots of the log-magnitude and the phase, using log-frequency in rad/s as the ordinate.
- (c) Make a polar plot of the frequency response.

Q2. Draw the Bode plots for the system

$$G(s) = \frac{K(s+3)}{s(s+1)(s+2)}.$$

$$G(s) = \frac{(s+20)}{(s+1)(s+7)(s+50)}.$$